Questions by topic: Energy conservation: transport, buildings, industry - Set 1 (80 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows the energy used per passenger-kilometre by different forms of transport when carrying a typical load.

Table 1

Mode of transport	Typical occupancy	Energy use / MJ per passenger-km
Car (petrol, 1 occupant)	1	2.9
Car (petrol, 4 occupants)	4	0.7
Bus (diesel)	25	0.6
Electric train	200	0.3
Bicycle	1	0.06
Short-haul aircraft	120	2.4

0 1 . 1 Calculate the percentage reduction in energy per passenger-km achieved by carrying four people in a car instead of one.

[1 mark]

Answer ☐ %

0 1 . 2 Explain why bulk transport (buses, trains) uses less energy per passenger-km than cars.

[2 marks]

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0 1 . 3 Suggest **three** reasons why people continue to use cars despite the energy advantages of public transport.

[3 marks]

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0 1 . 4 Explain how integrated transport systems and Active Traffic Management reduce energy use.

[4 marks]

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0 2 . 1 Explain how each of the following features of vehicle design reduces energy use.

[6 marks]

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Aerodynamics .....

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Reduced mass .....

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Regenerative braking (KERS) .....

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0 2 . 2 Explain why solid steel wheels on rails lose less energy than pneumatic tyres on roads.

[2 marks]

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0 2 . 3 Describe **two** ways combustion efficiency in an internal combustion engine can be improved.

[2 marks]

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0 3 . 1 Explain what is meant by the **embodied energy** of a vehicle and how it can be reduced.

[3 marks]

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0 3 . 2 Describe **four** principles of designing vehicles for end of life.

[4 marks]

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0 3 . 3 Explain why there is an optimum speed for fuel efficiency in a car.

[2 marks]

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0 3 . 4 Give **one** way driver behaviour or vehicle systems can reduce fuel use in congested traffic.

[1 mark]

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0 4 Table 1 shows the rate of heat loss through 1 m^2 of different glazing systems for a 1 °C temperature difference (U-value).

Table 1

Glazing	Heat loss / W m^{-2} K^{-1}
Single glazing	5.8
Double glazing, air filled	2.8
Double glazing, argon filled, low-e glass	1.6
Triple glazing, krypton filled, low-e glass	0.8
Double glazing, vacuum (0.2 mm gap)	0.74

0 4 . 1 A house has 20 m^2 of single glazing. The average temperature difference between inside and outside over the heating season is 12 °C. Calculate the reduction in the rate of heat loss if the windows are replaced with argon-filled low-e double glazing.

[3 marks]

Answer ____ W

0 4 . 2 Explain why the gap between the panes should be as large as possible but not too large.

[2 marks]

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0 4 . 3 Explain how low-emissivity glass and inert gas filling each reduce heat loss.

[2 marks]

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0 4 . 4 Explain why vacuum double glazing has a very narrow gap and where it is used.

[2 marks]

0 4 . 5 State **one** reason why insulation thickness has an optimum value.

[1 mark]

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0 5 . 1 Explain how each of the following building design features reduces energy use.

[6 marks]

Orientation

Low surface area : volume ratio

Earth sheltering

0 5 . 2 Compare the embodied energy of concrete with alternative building materials.

[2 marks]

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0 5 . 3 Describe how a heat exchanger allows a building to be ventilated with minimal heat loss.

[2 marks]

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0 6 . 1 Explain how occupancy sensors and programmable thermostats reduce energy waste.

[2 marks]

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0 6 . 2 Explain why LED lights use much less electricity than the filament and CFL lamps they replaced, and give **two** other examples of low-energy appliance design.

[3 marks]

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0 6 . 3 Describe **three** changes in human behaviour that reduce domestic energy use.

[3 marks]

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0 6 . 4 Explain why heating water on demand saves energy compared with storing hot water.

[2 marks]

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0 7 Figure 1 compares the energy flows in a conventional gas-fired power station and a combined heat and power (CHP) station, each supplied with 100 units of fuel energy.

Figure 1

	Conventional power station	**CHP station**
Electricity output / units	40	35
Useful heat supplied to buildings / units	0	45
Waste heat lost / units	60	20

0 7 . 1 Calculate the overall efficiency of each station.

[2 marks]

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Answer [ ] %

0 7 . 2 Explain why the electrical efficiency of a CHP station is often deliberately kept below 40%.

[2 marks]

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0 7 . 3 Explain why CHP stations must be located close to the users of the heat.

[1 mark]

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0 7 . 4 Explain how each of the following features of an industrial heat exchanger increases its efficiency.

[3 marks]

Long narrow copper pipes .....

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Counter-current flow .....

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Reduced flow rate .....

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0 7 . 5 Explain why a single large spherical tank loses less heat than several small cubic tanks holding the same total volume of hot fluid.

[2 marks]

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0 8 . 1 Explain how integrated manufacture saves energy in the iron and steel industry.

[2 marks]

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0 8 . 2 Explain why recycling is not always energy efficient.

[2 marks]

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0 8 . 3 Explain how each of the following manages electricity infrastructure to conserve energy.

[4 marks]

High-voltage grid .....

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ICT management of supply and demand .....

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0 8 . 4 Give **two** locational factors affecting where new generating infrastructure is built.

[2 marks]

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END OF QUESTIONS

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